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Transforming Real Time Intelligence: A Case Study on Multidisciplinary Integration Through Collaborative Planning Environment in Onshore Abu Dhabi Fields

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Abstract

Drilling anomalies continue to occur across well operations despite the availability of extensive relevant data. This gap between available knowledge and execution outcomes highlights the limitations of traditional planning approaches and the difficulty in effectively leveraging historical data in real time decision-making. The integration between collaborative multidisciplinary planning software and a knowledge management system presents a breakthrough in planning wells. This case study explores benefits of the integrated environment, focusing on the enhancement operational efficiency.

The integration between the collaborative well planning software and the Knowledge Management (KM) system provides a seamless digital connection between the multidiscipline engineering departments and the rig sites. This allows the asset teams to better visualize and manage data improving operational decisions in real time. The integrated environment uses graphic symbols to represent different downhole incidents and display them in 3D subsurface models and wellbore views. The integrated software enables a diagnostic review of stored data, displaying incidents and their exact coordinates on the 3D subsurface model. This allows studying the relationship between incidents, their frequency, and location.

The Knowledge Management workflow through the integrated software generates high quality drilling and completion historical datasets and lessons learned which are utilized in future projects to optimize well plans and consequently improve drilling performance. The integrated software helps diagnose planned and unplanned activities, non-productive durations, and other key depth-based data. Comprehensive 3D visualization helps study how drilling incidents relate to known faults and surface grids, and where certain incidents tend to occur. This visualized correlation helps improve the decision-making process for the multidisciplinary teams. The integrated software also serves as a central hub for drilling-operations knowledge management. This enables reviewing potential incidents, preparing contingency plans, and capturing lessons learned. Collaboration between drilling, geoscience, petrophysical, and assets teams is significantly optimized through the collaborative environment presented by the integrated software. Real time data feeding into the integrated software coupled with the 3D digital visualization permits a wholistic management of the operations across the field, resulting in the following benefits:

- Stuck pipe depths analysis
- Mud loss identification
- Reduction of geo-mechanical uncertainty
- Sophisticated sidetrack kick-off points analysis
- Trajectory complexity analysis
- Reaming events analysis
- Reduction of non-productive time
- Improved real time interventions

The successful coupling between a collaborative multidisciplinary planning software and a knowledge management system is newly trialed at the onshore fields of Abu Dhabi. Testing the true efficiency of the studied collaborative integrated digital environment on a wide range of multidisciplinary data and operations provides novel results. This unlocks hidden knowledge that is efficiently utilized to optimize well planning, operations, and associated costs.

Introduction

In recent years, thousands of oil and gas wells have been drilled in UAE Onshore brown fields. At all fields, an extensive amount of historical data has been accumulated and stored in Master Data Storages (MDS) of the Company. This information consists of high-frequency data through the RTOC and low-frequency data through the daily drilling reports. The proposed solution offers to combine this data in a collaborative and integrated visualization environment. It further contextualizes this information by subsurface geological and geo-mechanical models to understand the dependencies of different drilling events from wellbore conditions and come up with better decisions during future well planning and well monitoring activities ([Pathy et al, 2016](#)).

Currently, the drilling process operates in a predominantly reactive manner, with adjustments typically made only after anomalies or challenges are observed. This sometime can lead to critical consequences and Non-Productive Time (NPT) that results in the loss of valuable rig durations and resources. Effective management of drilling procedures requires adopting both proactive and preventive approaches to operational challenges. It is confirmed that noticeable changes in drilling parameters can be observed while bit penetrates different formations with different lithology and respective densities. Contextualizing drilling data by subsurface models gives an opportunity to predict surface data logging parameters by correlating these parameters from historical data justified by changes in geological conditions ([Roseland et al, 2010](#)).

The subsurface models contain critical data like formation tops, faults, and 3D distribution of key geo-mechanical data such as pore pressure, fracture gradient, orientation of horizontal stresses, and unconfined compressive strength. Drilling data on the other hand contains well logs, surface data logging parameters such as hookload, weight on bit, rotation of drill string, torque, standpipe pressure, mud flow, rate of penetration, and description of critical drilling events such as stuck pipe, well control, and mud losses ([Marir et al, 2015](#)).

This paper reviews a novel approach of deploying a solution at Abu Dhabi onshore fields to enhance and optimize well construction processes and find a way to avoid undesirable critical drilling events and NPTs.

Concept and Scope

The Challenge

The primary challenge currently faced in optimizing well construction and drilling operations stems from the massive volumes of data accumulated over many years of company drilling activity. While this extensive historical record offers valuable insights, it has become increasingly difficult to analyze quickly and extract actionable recommendations in real time. Predictions of Non-Productive Time (NPT) is not always satisfactory; a closer examination of the data reveals that many opportunities exist to transition from a reactive approach to a more proactive strategy. Achieving this shift, however, requires deeper and more comprehensive datasets—ones that extend beyond the real-time data typically transferred and monitored. By integrating a richer array of historical and contextual information, the company can better anticipate challenges, reduce NPT, and enhance the overall efficiency of drilling operations.

Objective

The primary objective is to develop a comprehensive tool capable of efficiently displaying and rapidly analyzing the diverse, multidisciplinary data collected over years of well operations, thereby enhancing the real time monitoring of wells. Additionally, the solution aims to predict and proactively address Non-Productive Time (NPT), minimizing unnecessary delays and equipping teams to effectively manage and reduce any potential loss time during drilling activities. This will not only improve the performance of ongoing operations but also support achieving the grand wells delivery mandates of all fields.

Concept

A multidisciplinary software solution has been selected and deployed to integrate a wide variety of data sources, including subsurface information such as surfaces, faults, 3D grids, and models, as well as historical drilling data like daily drilling reports and performance records. Additionally, the system incorporates both archived real-time and ongoing drilling parameters and logs. This comprehensive integration enables users to quickly and easily visualize potential issues at the field level and in offset well areas, thereby facilitating more informed decision-making and proactive management throughout the well construction process.

Scope

In a conventional working environment, geoscientists, reservoir engineers, and drilling engineers work on projects separately. The workflow between these disciplines is typically challenging and time consuming. The lengthy well planning process results in losing millions of dollars due to unoptimized development plans. It may also lead to potential errors or missing critical data.

The proposed software technology, a unified visualization, interpretation, and modeling workspace where asset teams can collaborate more effectively to evaluate and develop assets. The software delivers a true multi-user environment with unprecedented integration across multi-domain workflows and data types. It represents a combination of automation and patented optimization technology. It reduces planning durations from months to days and in some cases reduces cycle times by up to 80% in comparison with conventional workflows. Well Planners can generate well trajectories, pad or platform locations, and reservoir targets in the 3D visualization environment. Asset teams can generate and evaluate more development scenarios and quickly respond to risks or changes to scheduling or capital expenditure constraints. The figure below demonstrates a collaborative and dynamic environment where different data types from different teams and workflows connected together to achieve the most optimized well plans.

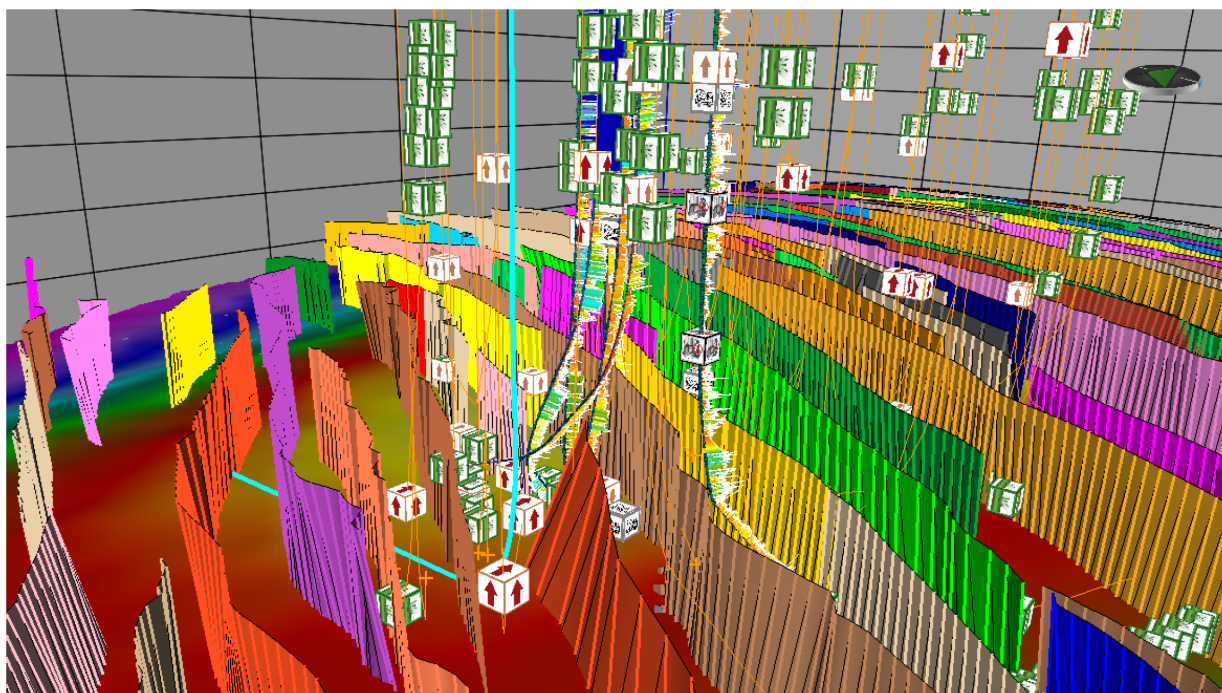


Figure 1—Software collaborative environment

Solution Description

The key process in delivering Knowledge Management (KM) workflow connected to real time data is relevant and comprehensive data management at all phases in well construction cycle. There are 4 main phases in well construction life cycle: Plan, Design, Execute and Review. Please see the figure below illustrating the main phases.

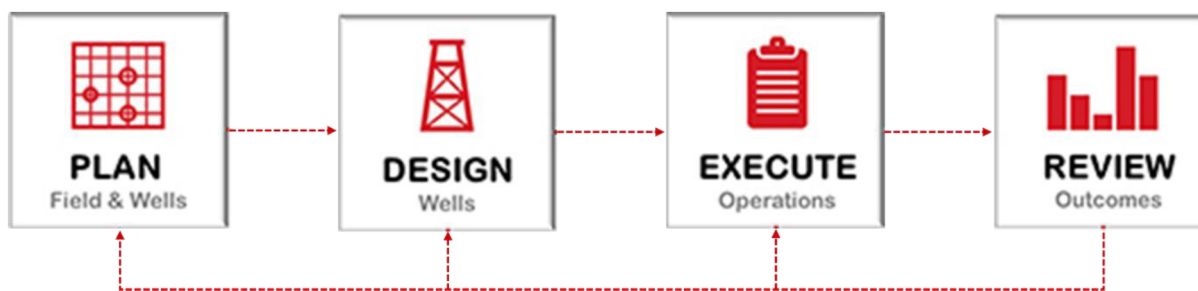


Figure 2—Well construction life cycle

For the most optimum and reliable results in well construction process all represented phases should be interconnected through data flows. Multiple data types from different domains are involved in this process (Reddicharla et al, 2017):

- Well trajectories and anti-collision models
- Casing data
- Daily drilling reports (DDR) and lessons learnt
- Surface data logging parameters: hook-load, weight on bit, rotation of drill string, torque, standpipe pressure, mud flow, and rate of penetration
- Well logging data: Gamma ray, resistivity, acoustic, and sonic data

- Formation and reservoir models in 2D and 3D formats including faults
- Seismic in 2D and 3D formats

To achieve the fastest generation of risk management and mitigation plans the mentioned data types should be in high quality and properly stored in the structured Master Data Storages (MDS). Assigned data management personnel is required to support maintaining data quality and continuous data streaming updates. The following are the key examples of what data management process looks like:

1. Data management for lessons learnt: the specialist should compile a set of critical drilling historical events (also known as lessons learnt) generated from DDRs. These lessons can describe drilling events such as stuck pipe, well control, mud losses, tight holes, washouts, caving, casing shoe depths and others. One challenge occurs due to a high number of errors in the interpretation of these events. The data management process should involve validation and confirmation of such lessons. A data management specialist should merge the data from DDRs with surface data logging parameters and match it with subsurface models for validations. Only after this can the lessons be stored as reliable knowledge that is suitable for generating risk management and mitigation plans.
2. Data Management for Well Trajectories: the specialist should store actual trajectory surveys and planned well path at master data storage that is connected to the solution by defined integrated layer. Each well trajectory should be assigned to survey error tool model for anti-collision purposes. This will enable well trajectories to be automatically extracted for consumption in collaborative visualization environment.
3. Data management for actual casing shoe locations: the common procedures for defining casing shoe locations follows assignment of actual hole depths for each section to the nominated casing shoes. The specialist should monitor the process of casing running operation and adjust casing shoe locations according to actual casing string length. This will enable analysis of casing shoe placement in accordance with formation tops.
4. Data management for surface data logging parameters: the specialist should operate with high-frequency data storage to qualify and validate the parameters to be consumed by the Solution. This will give an assurance of excluding any mistakes while contextualizing surface data logging parameters by subsurface models.
5. Data management for formation and reservoir models: The solution includes a master data storage for subsurface models, which should be carefully migrated from various company data sources to ensure they are ready for use by the solution.

Figure 3 illustrates the detailed structure of the proposed solution, the collaborative working environment.

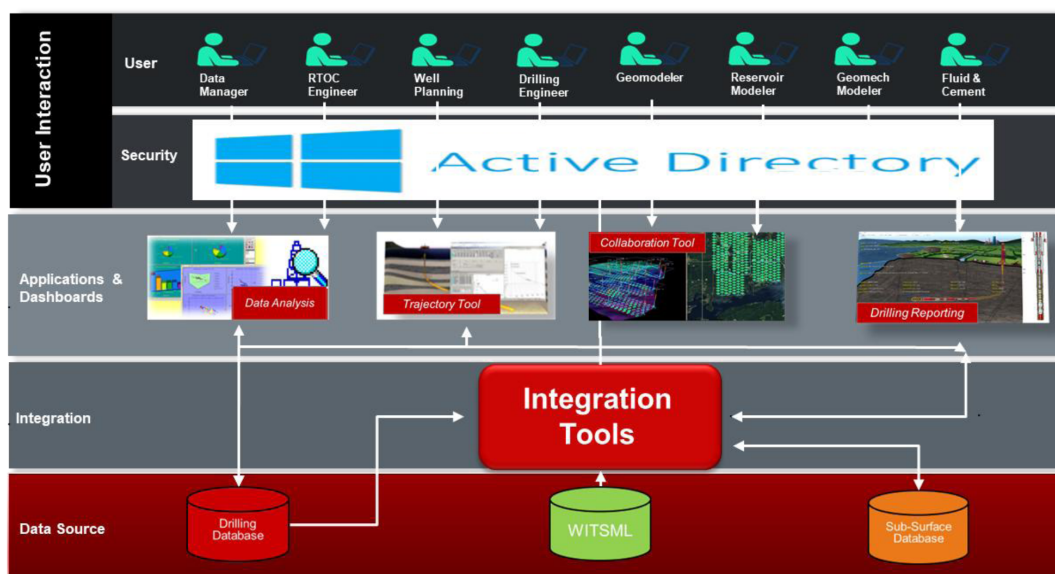


Figure 3—Solution architecture

Results and Discussion

There are outstanding results derived from the implementation of the presented data management solution. The impact of these results is also examined within the collaborative working environment. By evaluating key performance indicators and operational outcomes, the discussion highlights how structured data storage, validation protocols, and integration of diverse datasets contribute to enhancing risk management, decision-making, and consequently achieve major company mandates.

Losses Prediction when Crossing Faults

A conventional method for predicting losses when planning a new well involves selecting offset wells within a radius of several kilometers from the target well. The process typically requires specialists to manually examine DDRs for each of these wells, extracting information about recorded loss events in order to assess whether the targeted area presents a higher risk of drilling fluid losses. While this manual review offers some insights, it is time-consuming and often fails to incorporate available subsurface data, such as geological models or formation properties, which can be critical for accurate predictions.

Importantly, when loss events from various wells are visualized within a three-dimensional (3D) environment, it becomes evident that there can be significant variability in loss severity even among wells located in close proximity. Some wells might exhibit minimal or no losses, while others may have experienced severe losses resulting in considerable NPTs and operational challenges. This spatial understanding, enabled by integrating losses data into a collaborative 3D visualization, provides a much clearer picture of risk distribution and enables a more focused approach to planning mitigation strategies.

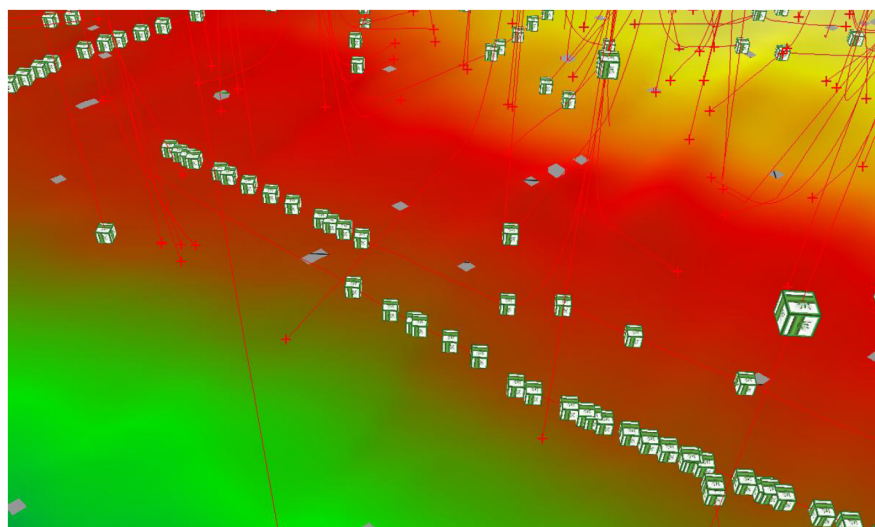


Figure 4—Mud losses events with formation tops only

When faults are analyzed in conjunction with DDR events, it becomes immediately apparent that every recorded loss incident within the examined area is directly correlated with fault intersections. This insight allows for a more precise and proactive risk management strategy. Specifically, it means that significant drilling fluid losses can be anticipated only in locations where the well trajectory crosses a fault. In practical terms, if a planned well path is projected to intersect a fault, the drilling team can make advanced preparations such as mobilizing additional chemicals, securing extra water supplies, or arranging for cement trucks. That is, to effectively mitigate potential loss events. Conversely, for well trajectories that do not intersect any identified faults, there is no need to commit these extra resources, as the likelihood of experiencing substantial losses is minimal. This targeted approach not only streamlines operational logistics but also reduces unnecessary expenditure, making the drilling process more efficient and cost-effective.

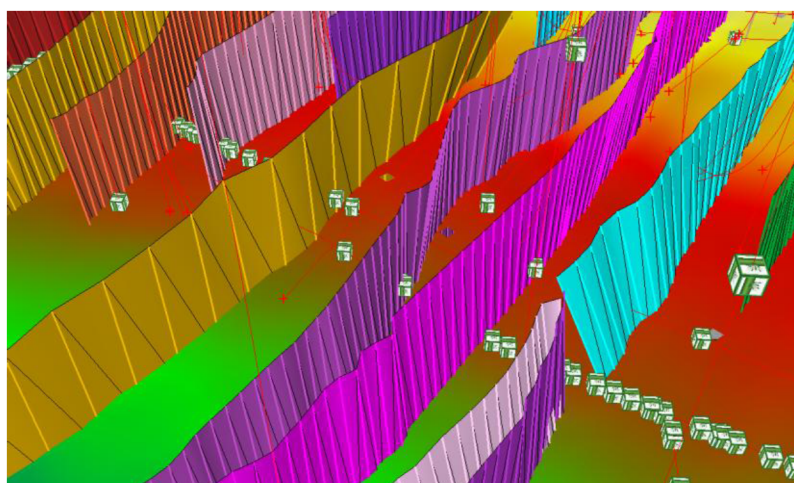


Figure 5—Mud losses events with formation tops and faults

In figure 6, the actual well encountered significant drilling fluid losses, which necessitated the installation of several Lost Circulation Preventers (LCPs) to control and mitigate these losses. By analyzing the planned well trajectory (highlighted in blue), it becomes possible to forecast with greater accuracy where substantial losses are likely to begin—specifically, these losses are expected to occur after the well crosses the identified fault line. This predictive insight enables the drilling team to take proactive measures by mobilizing a cement truck and other necessary resources in advance. Such planning is crucial for minimizing Non-Productive

Time (NPT) that would otherwise be associated with loss incidents. In this way, the integration of geological and operational data not only improves the precision of loss predictions but also enhances operational efficiency and cost-effectiveness throughout the drilling process.

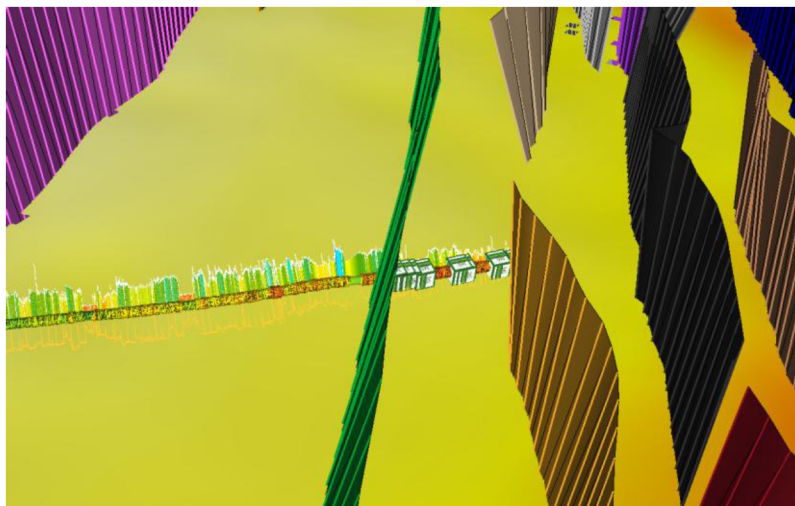


Figure 6—Mud losses events occurred after fault intersection

Identifying Hazardous Stuck Zones

Through comprehensive analysis and collaborative investigation of stuck pipe incidents, specific zones were identified where occurrences of stuck pipe were significantly more frequent and densely concentrated compared to other areas. Upon further examination, it became evident that every recorded stuck event in these high-risk zones was attributed to differential sticking. This discovery highlighted the necessity of reassessing the Pressure Profile–Mud Weight (PP-MW) margin, as well as conducting an additional review of Pore Pressure (PP) and Fracture Gradient (FG) values to ensure their accuracy and suitability for the planned operations. Based on these findings, the zones with elevated stuck pipe risk have now been formally designated as hazardous areas to be avoided in future collaborative well planning scenarios. By proactively flagging these problematic sections, operational teams can develop targeted mitigation strategies and improve the overall safety and efficiency of drilling campaigns.

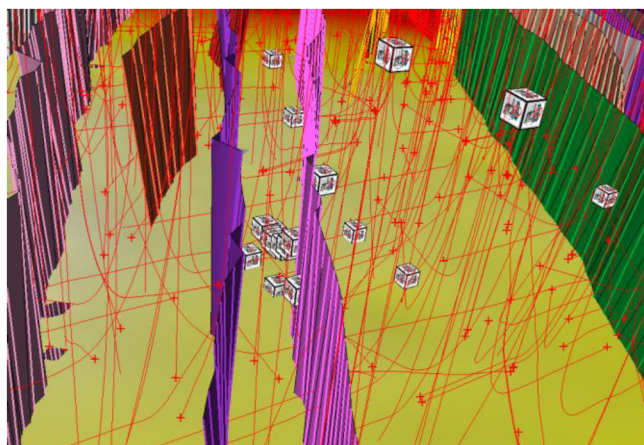


Figure 7—Hazardous areas with high risk of stuck pipe

Prediction of Low ROP Intervals

Incorporating a subsurface geo-mechanical model into the workflow allows for a comprehensive evaluation and prediction of intervals where low Rates of Penetration (ROP) are expected during drilling operations. By leveraging this model, the drilling team can proactively distinguish between ROP declines that are a natural consequence of geological factors. This includes sudden increases in unconfined compressive strength (UCS) of the formation. It is also affected by operational inefficiencies, such as bit balling or suboptimal drilling parameters like rotations per minute (RPM), weight on bit (WOB), or flow rate.

When both the geo-mechanical model and offset well data indicate that a drop in ROP coincides with an abrupt change in UCS, it becomes clear that the slowdown is geologically driven. In such cases, unnecessary trips out of the hole and equipment inspections can be avoided, saving both time and resources. Conversely, if there is a noticeable decrease in ROP but the UCS remains constant according to the model, this discrepancy triggers an alert for the Drilling Supervisor and Drilling Engineer. They are then prompted to investigate the underlying cause of the reduced drilling efficiency, which may include optimizing drilling parameters or performing a trip out of the hole to inspect the bit for potential damage.

This integrated approach not only enhances decision-making by distinguishing between geological and operational causes of low ROP but also improves operational efficiency. Ultimately, it leads to more effective drilling campaigns by minimizing non-productive time, preventing unnecessary interventions, and ensuring timely responses to real equipment or procedural issues.

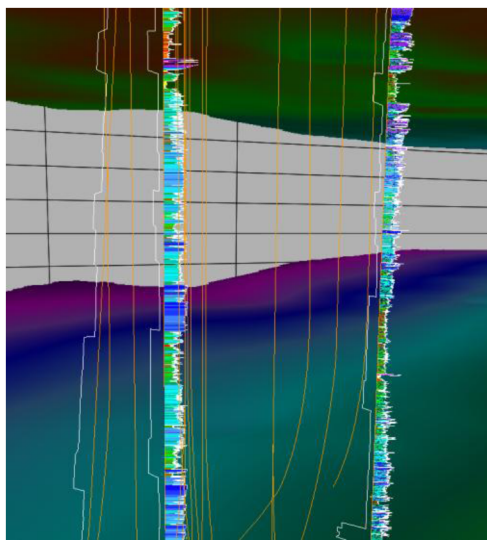


Figure 8—Unconfined compressive strength and ROP relation

3D Losses Map

By integrating subsurface geologic and geophysical (G&G) data with daily drilling reports (DDR), a comprehensive three-dimensional (3D) model for tracking and analyzing drilling fluid losses was developed. The process involved several detailed steps to ensure accuracy and usability:

1. Import in collaborative wellplanning environment actual trajectories, surfaces (formation tops), faults and losses KM events (from DDRs).
2. Finding interception points of all wells with surfaces and faults.
3. Export coordinates of interception points to PowerBI software to merge with information about losses from DDR.
4. Updating text description of each losses event with formation/fault name where losses occurred and loading updated KMs in collaborative well planning environment.

This robust, integrated database now provides operational teams, such as mud engineers and well planners, with rapid access to critical information regarding the most hazardous zones for fluid losses associated with each formation. Furthermore, the system enables automated analysis of patterns and dependencies, such as the correlation between fluid losses and fault crossings, thereby supporting more informed decision-making and enhanced risk mitigation strategies during drilling operations.

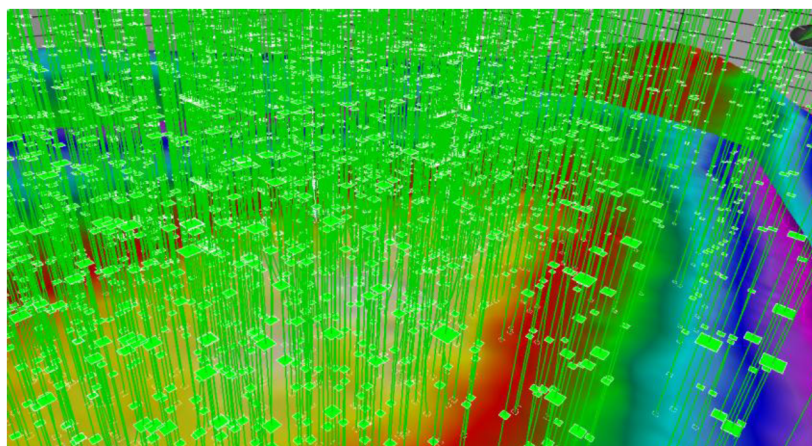


Figure 9—Formation intersections used in building 3D losses map

Collaborative Well Planning Sessions before Spudding the Well

The integration of huge set of data, including subsurface geology, real-time logging, and DDR records within a single collaborative environment has made it possible to conduct highly effective well planning sessions prior to the commencement of drilling. These sessions bring together drilling supervisors, drilling engineers, and other key operational team members to collectively review the comprehensive geologic and operational context in which the upcoming drilling will occur.

During these collaborative meetings, the environment is carefully showcased using data visualizations and cross-referenced information from offset wells, formation characteristics, and historical drilling events. The main objective is to identify and emphasize anticipated hazards and zones of concern, leveraging insights drawn from the combined data sources. By highlighting potential problem areas, such as sections with a history of significant fluid losses or complex fault crossings, the team is able to align on mitigation strategies and make informed decisions regarding drilling techniques and operational adjustments.

The objective of these well planning sessions is to proactively reduce NPTs, enhance operational safety, and improve the overall efficiency of the drilling process. Through collaborative review and planning, all stakeholders are better prepared to address challenges as they arise, thereby increasing the likelihood of a successful and incident-free well.

Observations

Currently, the Company is setting up the Projects to run the Solution for different oil and gas fields following data management rules and requirements. Below is some information regarding data content for one of the brown fields:

Table 1—Project Statistics

Overall amount of the wells	Mud Loss Events	Stuck Pipe Events	Well Control Events
More than 2000	~ 10000	~ 700	~ 40

The Company will be benefiting from Solution by targeting reduction in Geological NPT and gaining experience from similar Projects in Norway where the same type of the Solutions has been deployed and significant improvements has been observed. This case study shows that common visualization and multi-discipline team working to ensure the delivery of integrated mitigation plans and contingencies which are focused and relevant, with emphasis on the major and most likely risks making the most impact on geologically related NPT reduction. This has been borne out when comparing the dramatic reduction in NPT on the well #2 (9.4% of Geological NPT in comparison with well #1 (20 % of Geological NPT) and the further improvement on well #3 (3% of Geological NPT) (C. Telford et al, 2004). Table 2 summarizes the main achievements from this Project.

Table 2—Norway Project achievements

Well	Geological NPT	well-by-well % of reduction	Comments
Well #1	20%	-	Solution is NOT in Action
Well #2	9.4%	53%	Solution in Action
Well #3	3%	68%	Solution in Action

This Project does not require additional manpower to be allocated and assigned separately as Company's drilling engineering competency is sufficiently covering all the aspects required to run the Solution. At the same time, it is worth mentioning the need in organizing collaborative day-by-day sessions to review current progress and deliver informed and timely decision making. Collaboration and multi-discipline integration at the planning stage, coupled with the ability to monitor the well construction real-time from multiple locations, is proven to be an invaluable tool for improving overall operational efficiency. Drilling Engineers, Rig Superintendents, Geological and Geophysical groups are in the position to make fast and sound decisions, whereby they can communicate promptly with rig representatives on drilling issues, bore hole conditions and correlation of LWD logs against proposed formations tops. (EeVern Tan et al, 2007)

The main bottleneck observed during the Project is identification of genuine depths points of the event and contextualization of this event with subsurface models. Historically all the subsurface models have been converted from legacy Coordinate System to the latest one. This gave some distance shift while referencing models to the datums. The Company is extensively working on updating all subsurface models with correct assignments to the latest datums and Coordinate Systems aligned with wells placement. That's why the level of confidence for contextualization of drilling events with subsurface models will be improved as well.

Conclusion

In summary, the integration of comprehensive geological, operational, and real-time drilling data within a collaborative well planning environment represents a significant advancement in the management of subsurface hazards. By centralizing critical information, such as fluid loss events, fault intersections, and formation properties, and making it readily accessible to multidisciplinary teams, organizations can facilitate more informed and proactive decision-making. The resulting well planning sessions not only enhance the team's ability to anticipate and mitigate risks but also enhance communications and alignment among all stakeholders involved in drilling operations.

This approach leads to a measurable reduction in NPTs, improved safety, and increased operational efficiency. As the industry continues to embrace digitalization and data-driven methodologies, such integrated systems will become even more vital in optimizing well delivery and ensuring the successful execution of complex drilling projects.

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Nomenclature

KM:	Knowledge Management
MDS:	Master Data Storages
DDR:	Daily Drilling Report
NPT:	Non-Productive Time
LCP:	Lost Circulation Preventers
ROP:	Rate of Penetration
UCS:	Unconfined Compressive Strength

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